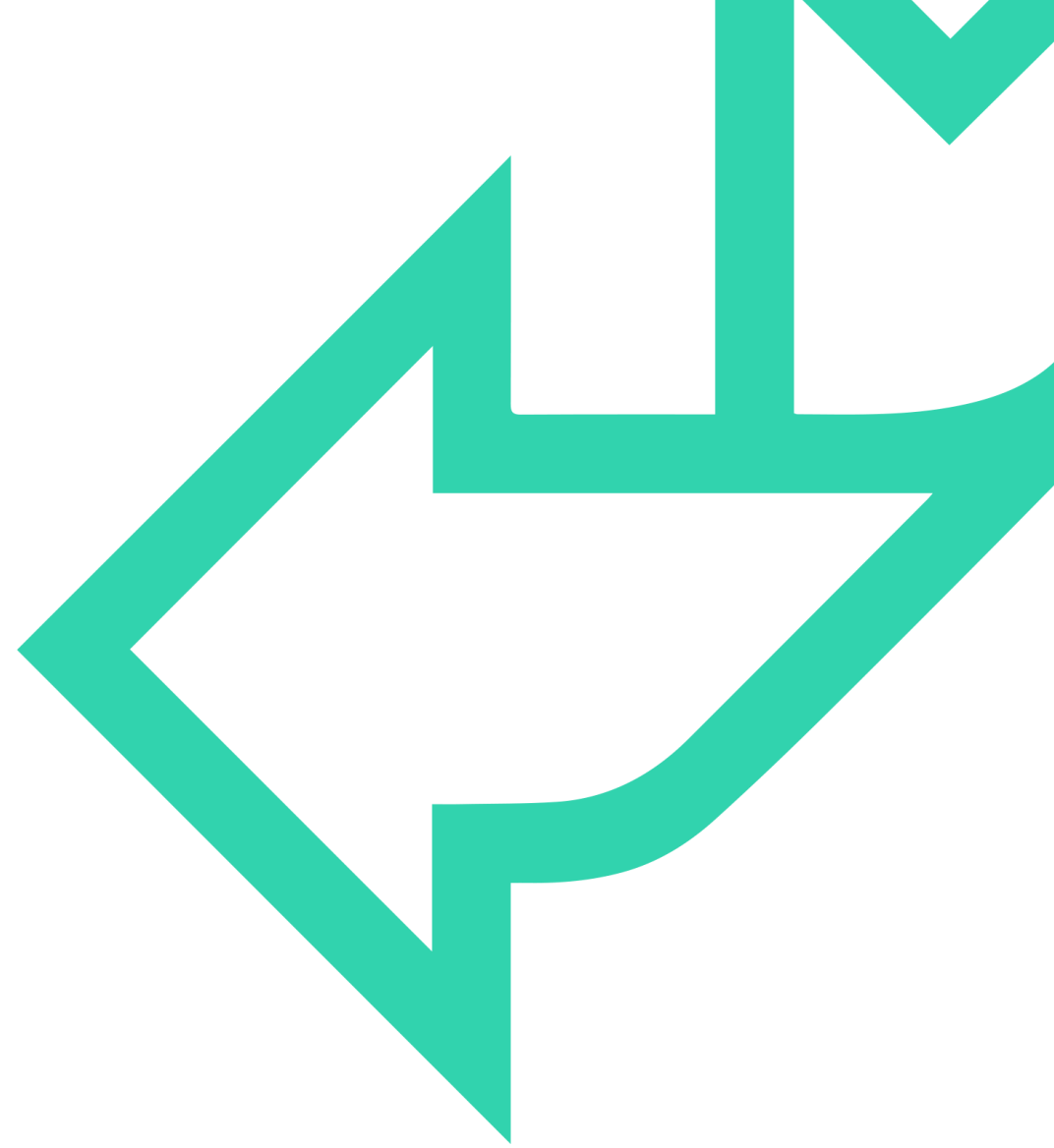


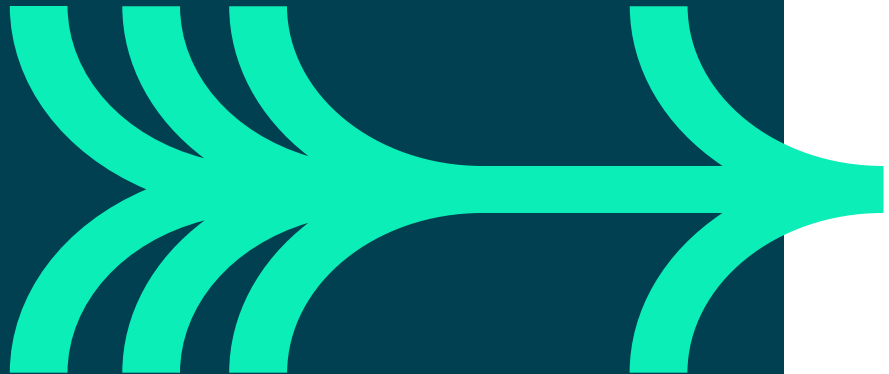


Exception Handling





WHAT IS EXCEPTION HANDLING?



- **Every operation involving user data entry may result in a crash**
 - Every I / O operation, files, databases, etc. may crash
- **Good programs are bullet proof**
 - Bad ones dump an error message and die!
- **It may not be possible to make safe every route through an app**
 - ...But you must try
- **Catch exceptions in functions or let them go up the calls stack**
- **When an exception is handled, it's cleared as if no error occurred**



EXCEPTION HANDLING EXAMPLE



```
import pyodbc

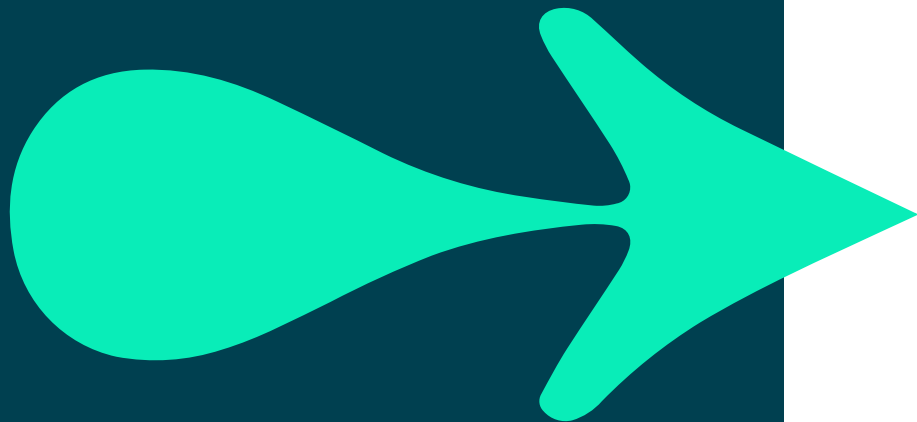
def showCompany():
    connectionString = r'DRIVER={ODBC Driver 13 for SQL Server};

                        SERVER=.\SQLEXPRESS;DATABASE=qastore;Trusted_Connection=yes'
    try:
        conn = pyodbc.connect(connectionString)
        cur = conn.cursor()
        result = cur.execute('SELECT * FROM companys').fetchall()
        conn.close()
        return result
    except:
        return None

#----- main -----
rows = showCompany()
if rows != None:
    for row in rows:
        print(row)
else:
    print('Error reading data.')
```



EXCEPTION HANDLING IN FUNCTION



```
import pyodbc

def showCompany():
    connectionString = r'DRIVER={ODBC Driver 13 for SQL
Server};
SERVER=.\SQLEXPRESS;DATABASE=qastore;Trusted_Conn
ection=yes'
    try:
        conn = pyodbc.connect(connectionString)
        cur = conn.cursor()
        result = cur.execute('SELECT * FROM
company').fetchall()
        conn.close()
        return result
    except Exception as ex:
        print("Error: ", ex)
        return None

#----- main -----
rows = showCompany()
if rows != None:
    for row in rows:
        print(row)
```



EXCEPTION HANDLING IN MAIN



```
import pyodbc

def getQASoreConnection():
    connectionString = r'DRIVER={ODBC Driver 13 for SQL Server};

                        SERVER=.\SQLEXPRESS;DATABASE=qastore;Trusted_Connection=yes'
    conn = pyodbc.connect(connectionString)
    cur = conn.cursor()
    return conn

def getQASoreRows(sql):
    try:
        conn = getQASoreConnection()
        cur = conn.cursor()
        result = cur.execute(sql).fetchall()
        conn.close()
        return result
    except:
        return None

#----- main -----
try:
    rows = getQASoreRows('SELECT * FROM company')
    if rows != None:
        for row in rows:
            print(row)
except:
    print('Error reading data.')
```



Any questions?

A large, stylized teal arrow pointing to the right, composed of two horizontal bars and a triangular head, spanning the width of the slide.

Thank you!

Hope you enjoyed this learning journey!